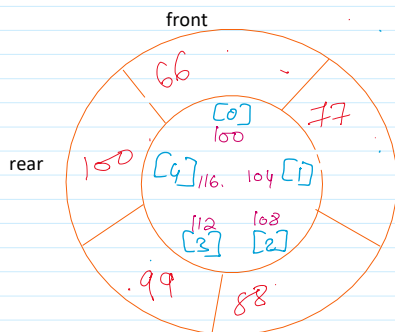


int arr[5]

=-1

Queue full condition

**Enqueue:**

- 1) Increment rear as
Rear = (rear+1) + SIZE
1. Add element at rear position
2. If front == -1, make front = 0

Dequeue :

- 1) Increment front.
- *****

If front is 4 and rear is 0
to delete the rear position, we cannot
increment front as front ++ Will be
index 5 but we want to delete index 0;
So,
Front = front + 1 % SIZE

If deleting the last element in queue
If (front == rear)
Front = rear = -1

Queue Empty condition

If (rear == -1) queue is empty

Queue Full condition :

When Queue is full,

Rear = 0 front 1
Rear = 1, front = 2
Rear = 2 front = 3
Rear = 3 front = 4
rear = 4, front = 0

This means,
Front == rear + 1

But when rear = 4 front = 0
0 == 4+1 does not satisfy the above condition
So the queue full condition can be
Front == (rear+1) % SIZE
0 == (4+1) % 5
0 == 5 % 5
0 == 0

Option 2:

If rear == SIZE-1
Rear = 0;
Else
Rear++;

$$\begin{aligned} \text{front} & & \text{rear} \\ \checkmark 1. & == (0+1) \% 5 \rightarrow 1 \\ 2. & == (1+1) \% 5 \rightarrow 2 \\ 3. & == (2+1) \% 5 \rightarrow 3 \\ \checkmark 4. & == (3+1) \% 5 \rightarrow 4 \\ \boxed{\text{front}} & == (\text{rear}+1) \% \text{SIZE} \\ 0 & == (4+1) \% 5 \rightarrow 0 \\ 0 & == 5 \% 5 \end{aligned}$$

$$\begin{aligned} \text{rear} &= (\text{rear}+1) \% \text{SIZE} \\ &= (-1+1) \% 5 \rightarrow 0 \% 5 \\ \text{rear} &= 0 \\ &= (0+1) \% 5 \\ &= 1 \% 5 \\ \text{rear} &= 1 \\ &= (1+1) \% 5 \\ &= 2 \% 5 \\ \text{rear} &= 2 \\ &= (2+1) \% 5 \\ &= 3 \% 5 \\ \text{rear} &= 3 \\ &= (3+1) \% 5 \\ &= 4 \% 5 \\ \text{rear} &= 4 \\ &= (4+1) \% 5 \\ &= 5 \% 5 \\ \text{rear} &= 0 \end{aligned}$$

$$\begin{aligned} \text{front} &= (\text{front}+1) \% \text{SIZE} \\ &= (4+1) \% 5 \\ &= 0.5 \% 5 \end{aligned}$$